

We thank all reviewers for their careful, constructive, and fair evaluations. We appreciate the recognition of the paper’s broad evaluation across nine decoder-only language models, coverage of multiple model families and scales, and complementary probing tasks.

The reviewers raised several common concerns regarding novelty, causal interpretation, robustness, and presentation. In response, we not only clarify these points but also conducted additional experiments, including representation intervention, contextual sentence evaluation, cross-model transfer, and expanded evaluation metrics, which further strengthen the empirical evidence while remaining fully consistent with the paper’s central claim.

Novelty and contribution (Reviewers Ey2g, ATRq, and wJEU)

We agree that the paper does not claim methodological novelty through a new probing architecture. Rather, its contribution is a controlled empirical characterization of word-frequency representations across nine contemporary decoder-only language models. Beyond showing that frequency is decodable, the paper demonstrates three consistent findings across model families and scales: (i) word frequency remains strongly decodable throughout the network; (ii) the relative separation between high- and low-frequency words is strongest in the earliest layers and gradually decreases with depth; and (iii) strong absolute decodability persists in middle and late layers. This distinction between early relative separability and continued absolute decodability constitutes the paper’s primary empirical contribution.

Causal interpretation (Reviewers Ey2g, ATRq, and wJEU)

We agree that probing alone cannot establish how frequency information is causally used during model computation. Accordingly, the manuscript deliberately frames its conclusions as representational rather than functional or causal. Following the reviewers’ suggestions, we additionally performed representation-intervention experiments on Llama-3.2-1B and Qwen2.5-1.5B. Additive, subtractive, and nullification interventions produced only negligible changes in lexical-decision accuracy and the high-versus-low frequency gap, reinforcing that the recovered direction represents accessible information rather than a directly manipulable causal mechanism.

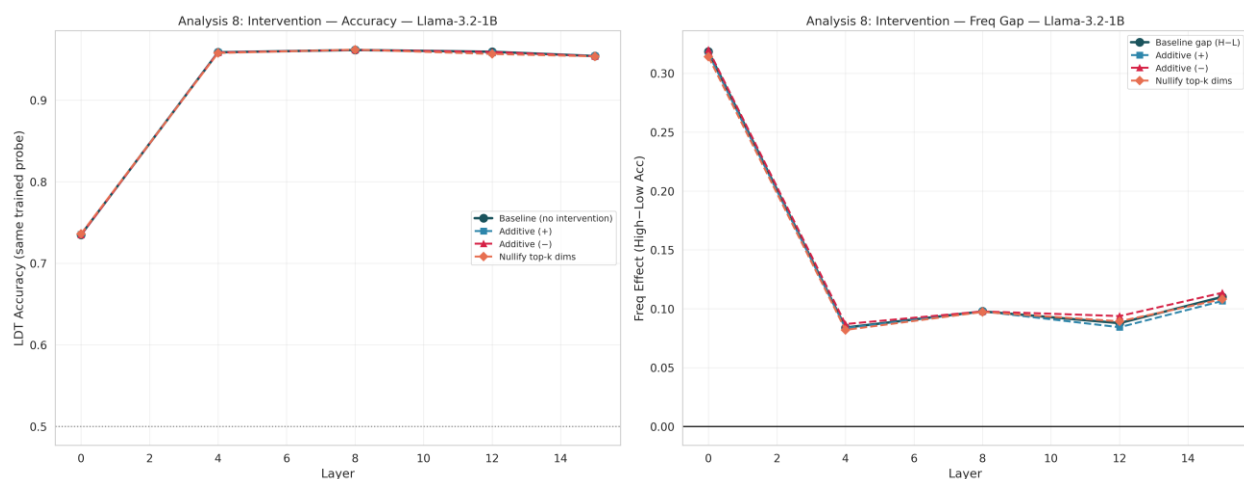


Figure 1. Representation intervention experiments. Simple perturbations of the recovered frequency direction produce minimal changes in lexical-decision accuracy and frequency separation, supporting our representational interpretation.

Alternative explanations and robustness (Reviewer ATRq)

Beyond the original robustness analyses—including tokenization-restricted evaluation, reduced-confound matching, pooling ablations, multi-seed stability, and probe-selectivity controls—we now additionally evaluate representation intervention, contextual sentence evaluation, and cross-model probe transfer. Collectively, these analyses demonstrate that the observed layer-wise pattern remains stable across substantially different evaluation settings and cannot be attributed to a single methodological artifact.

Scope and suggested extensions (Reviewers Ey2g and wJEU)

We appreciate the suggestions regarding contextual evaluation and cross-model probe transfer. Following these suggestions, we embedded target words into natural sentence contexts. Across both Llama-3.2-1B and Qwen2.5-1.5B, the same qualitative layer-wise trend is preserved: frequency remains highly decodable across the network while early layers exhibit the strongest relative separability. We also performed cross-model transfer experiments, which indicate partially shared representational geometry while still preserving architecture-specific characteristics.

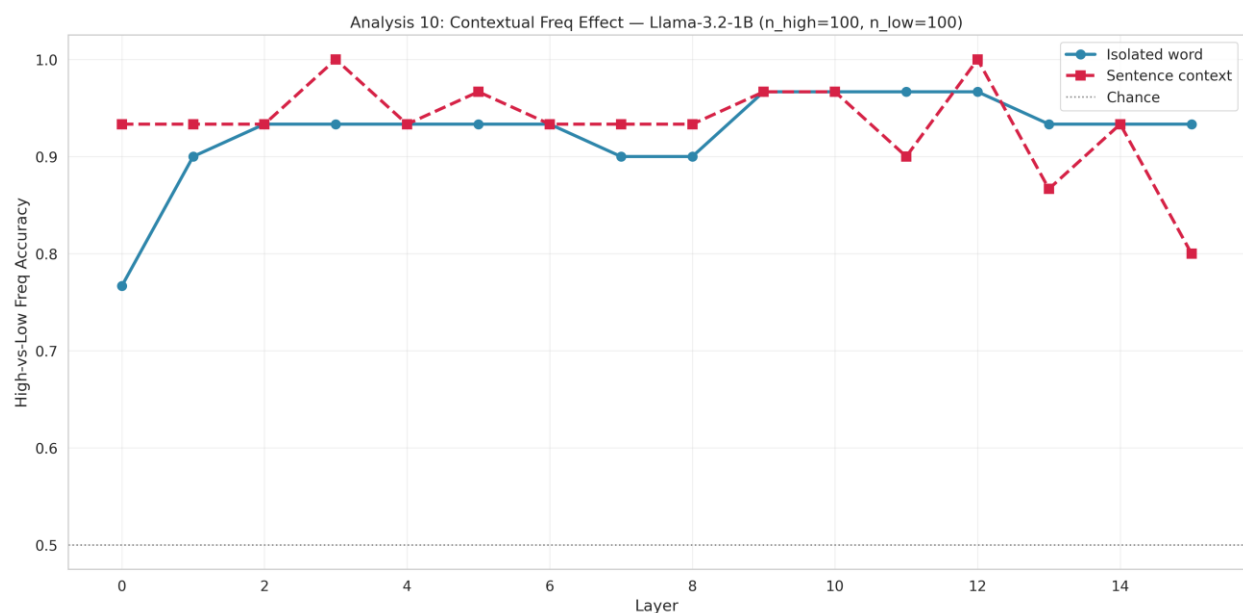


Figure 2. Contextual evaluation on Llama-3.2-1B. The qualitative layer-wise pattern remains consistent between isolated words and sentence contexts.

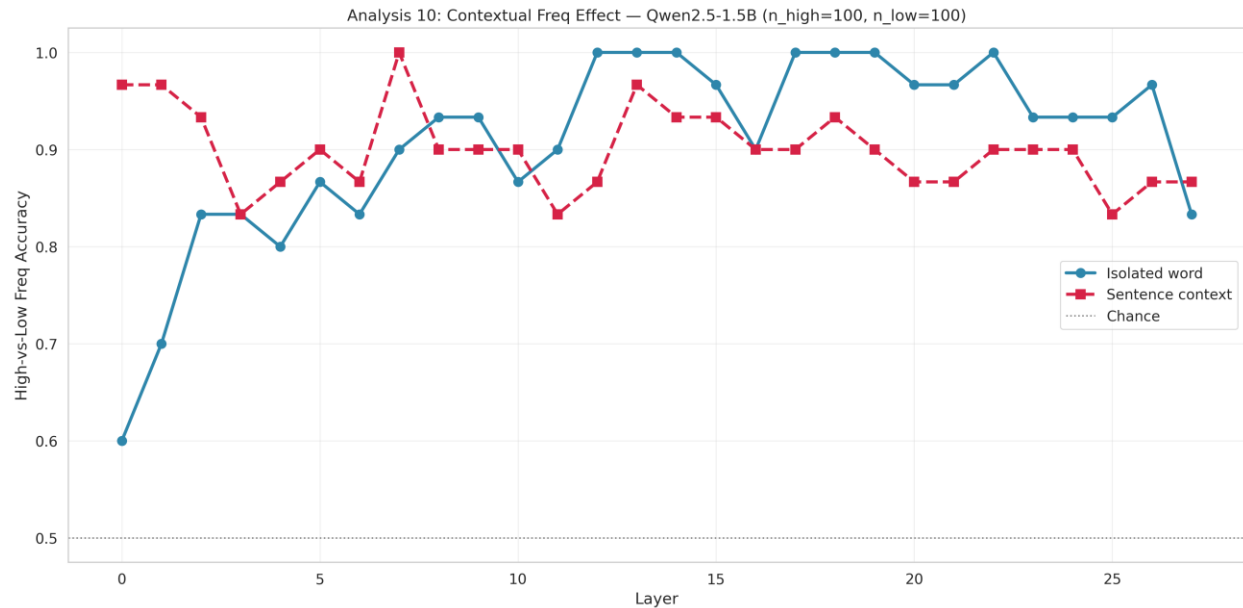


Figure 3. Contextual evaluation on Qwen2.5-1.5B. The qualitative layer-wise pattern remains consistent between isolated words and sentence contexts.

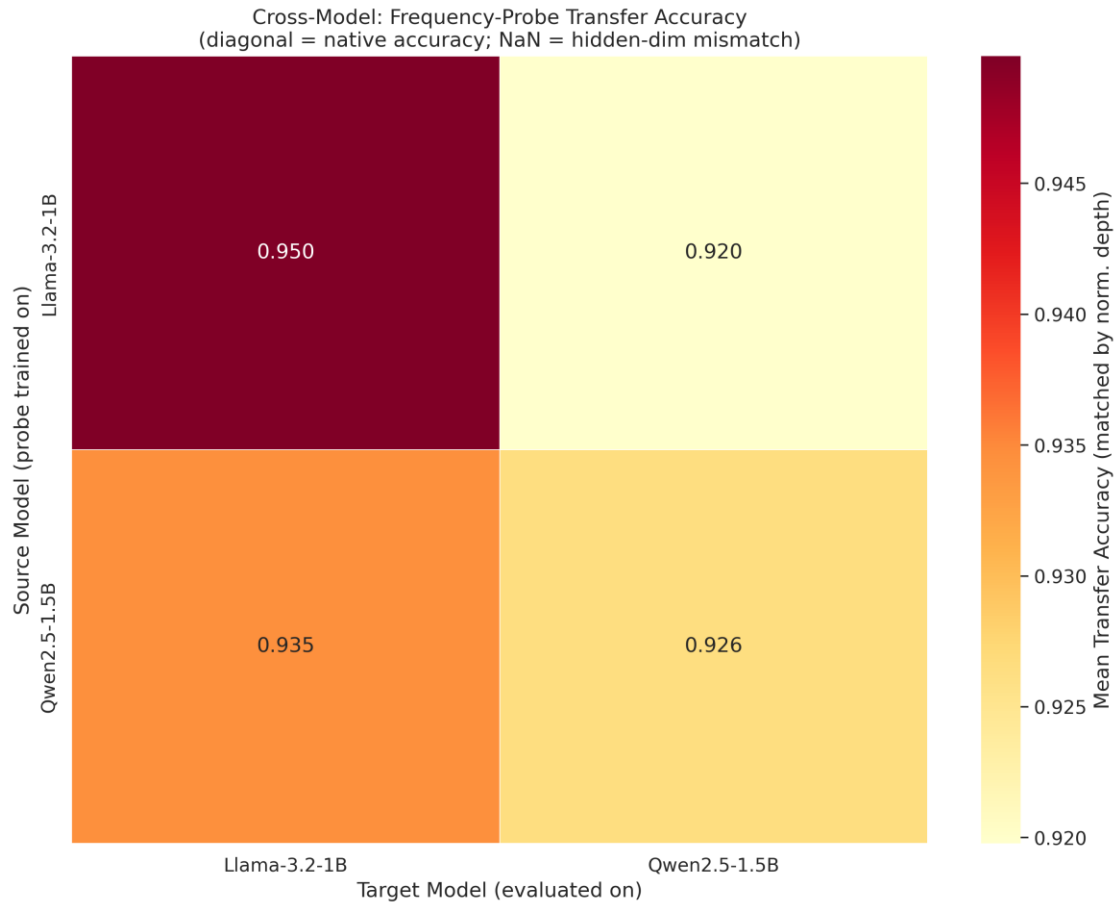


Figure 4. Cross model transferability confirms that a probe trained on the representations of one model generalize to the representations of another model.

Loss function and evaluation metrics (Reviewer wJEU)

We corrected the loss terminology: because $\gamma=0$, the optimization objective is weighted cross-entropy rather than focal loss. We also clarify that Table S4 reports F1 scores rather than accuracy. The direct binary frequency task is approximately balanced by construction, and the reduced-confound experiments further equalize group sizes. Threshold-free ROC-AUC values and explicit class distributions will be included in the revision.

Statistic	Value
High-frequency words	13,768
Low-frequency words	13,391
High-frequency mean tokens	1.83
Low-frequency mean tokens	2.67
High-frequency single-token (%)	29.39%

Low-frequency single-token (%) 0.57%

HAL frequency and lexical resources (Reviewer wJEU)

Frequency labels are derived from Log_Freq_HAL in the English Lexicon Project, which also provides the lexical-decision stimuli and psycholinguistic norms used throughout the study. This ensures that lexical items and frequency labels originate from a single internally consistent resource. We will clarify this rationale in the revised manuscript and discuss comparison with alternative resources such as SUBTLEX-US as an additional robustness analysis.

Methodological clarity and reproducibility (Reviewers ATRq and wJEU)

We will explicitly define HAL, lexical-decision norms, StandardScaler, ReduceLROnPlateau, controlled subset construction, MLP architecture and activation, BOS/prefix handling, the final non-padding token representation, and the anchor-layer sampling strategy. These implementation details are already reflected in our code and supplementary material but will be surfaced clearly in the main paper to improve reproducibility.

Length, organization, and presentation (Reviewers ATRq and wJEU)

We agree that the manuscript can be made more concise. Repeated methodological descriptions will be removed, discussion sections will be tightened without altering the scientific conclusions, Algorithm 1 will either be simplified or removed, notation will be clarified where retained, and minor typographical and formatting issues identified by the reviewers will be corrected.